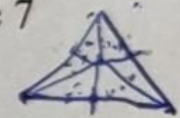
MST-II | **Subject: Discrete Mathematics (CPE-205) 2nd year/ 3rd semester** | **08/12/2021**

Note: *Section-A is compulsory. Attempt any two questions from Section-B* **MM.15**

Section-A (1 marks each)

1. Define isomorphism in graph with the help of an example.
2. What is the minimum number of NOR gate required to construct AND gate? Also construct it.
3. Draw 3-regular graphs with nine vertices.
4. Define Existential quantifier and Universal quantifier.
5. How many edges must be drawn in order to obtain a planar graph with 5 vertices that define 7 regions? Draw such graph. 10



Section-B (5 marks each)

6. State and prove the necessary and sufficient conditions for Euler circuits and paths.
7. **(a)** Construct the truth table for the compound proposition $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ **(2.5 marks)**
   (b) Let $= (\neg x \wedge z) \vee y$, write function f in disjunctive normal form (DNF). **(2.5 marks)**
8. Let G be the set of all positive rational numbers and * be the binary operation on G defined by $a * b = \frac{ab}{7}, \forall a, b \in G$, Prove that (G, *) is an abelian group.

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MM.15 Arnav Garg

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Subject: Discrete Mathematics Date 7 Nov 20?

Note: Section-A is compulsory & attempt any two questions from Section-B MST-II

B.Tech-Part-II 3rd sem) Branch: 2CE.

Section-A **(5*1=5 marks)**

Q1) a) Define Group with an example.

b) Draw a graph which is Hamiltonian but not Euler.

c) Define Conjunctive normal form (CNF).

d) Draw 3-regular graph using 6 vertices.

e) What is chromatic number of a graph?

Section-B (*Attempt any two*) **(5*2=10 marks)**

Q2) Explain Lagrange theorem.

Q3) Prove, using a truth table, that $\sim(p \rightarrow q) = \sim(\sim p \vee q)$

Q4) Show that the set $\{1, i, -i, -1\}$ forms a group under multiplication.

MST-II

Subject: Discrete Mathematics (CPE-205) 2nd year/ 3rd semester

06/11/2025

MM.15

Note: *Section A is compulsory. Attempt any two questions from Section B*

Section-A (1 mark each)

1. Define an isomorphism in a graph with the help of an example.
2. Define a homomorphism between two groups.
3. Draw 3-regular graphs with nine vertices.
4. Using Lagrange's Theorem, explain why a group of order 15 cannot have an element of order 6.
5. How many edges must be drawn to obtain a planar graph with five vertices that defines seven regions? Draw such a graph.

Section-B (5 marks each)

6. State and prove the necessary and sufficient conditions for Euler circuits and paths.
7. (a) Construct the truth table for the compound proposition $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$ **(2.5 marks)**

   (b) Let $= (\neg x \wedge z) \vee y$ Write function f in disjunctive normal form (DNF). **(2.5 marks)**
8. Let G be the set of all positive rational numbers, and * be the binary operation on G defined by $a * b = \frac{ab}{7}, \forall a, b \in G$, Prove that (G, *) is an abelian group.
9. Solve the Recurrence relation $a_n - 5a_{n-1} + 6a_{n-2} = 2^n + n$, given that $a_0 = 1, a_1 = 1$

Note: *Section-A is compulsory. Attempt any two questions from Section-B* **MM.15**

Kpattg 122

Section-A (1 marks each)

1. A graph has 21 edges, 3 vertices of degree 4 and other vertices are of degree 3. Find the number of vertices in the graph.
2. Differentiate between Hamiltonian and Euler graph.
3. Define pendant vertex, complete graph.
4. Differentiate between path and trail.
5. What is graph isomorphism? Give example.

Section-B (5 marks each)

6. State and prove the Lagrange theorem.
7.
   a. Prove that addition of two complex number $(a + bi)$ is an abelian group, where $a, b \in Integers$.
   b. Give an example of group which is not abelian.
8.
   a. Chromatic number of graph.
   b. Complete bipartite graph

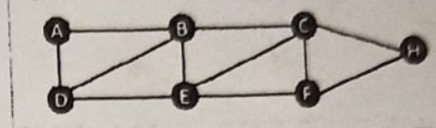
CSEB2305T: Discrete Mathematical Structures 12101078

MST-II/12/2022 Prem kumar

Max. Marks: 15 Time Allowed: 60 Minutes

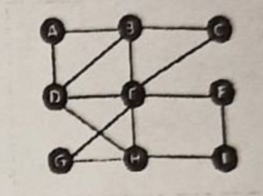
Section A is compulsory and attempt any *two* from Section-B

SECTION-A



Q1: Attempt all parts of this question.

i) Let G(V,E) is the given graph then find: →
   iii) all the simple paths from A to H ii) all cycle in the G
   iv) minimum distance(A, H) including vertices B and D 1



ii) Define Euler path. Prove that the following graph is Euler circuit or not. Is it traversable ? 1

iii) Let Q is a set of rational number and * is the operation defined on it by $a*b=a+b-ab$ then find out: 1
   ii) 3*4 ii) 2*(-5) iii) 7*(1/2)
   Also prove that (Q, *) is semigroup and is it commutative? 1

iv) Drawing truth table of the logical expression $(a \vee b) \wedge [(\neg a) \wedge (\neg b)]$, prove that if it is a contradiction or not ? 1

v) Prove that $\neg(a \vee b \vee c)$ and $[(\neg a) \wedge (\neg b) \wedge (\neg c)]$ are logically equivalence ?

SECTION-B

Q3 If Z_6 is the set of equivalence classes generated by the equivalence relation "congruence mod 6" then prove that $(Z_6, \times_6)$ is monoid. The operation $\times_6$ on Z_6 is defined as $[r]\times_6[c]=[(r\times c) \bmod 6]$ for any $[r], [c] \in Z_6$. 5

Q4 G(V,E) with $e=|E|$ and $v=|V|$ is a connected plannar graph with $e \geq 3$. Prove inductively that number of vertices V, number of edges E and number of regions R satisfy V-E + R=2 for G. 5

Q5 Define '*Hamiltonian circuit*' and '*Hamiltonian graph*'. 5
   iii) How many Hamiltonian circuits does a graph G with 5 vertices have? Justify your answer with example.
   iv) Does a Hamiltonian path or circuit exist on the given graph G →